



Ad_Hoc Insights Consumer Goods

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Why?

Objectives

What?

Company's detail and market

How?

Data, request and tools



Why?



Objectives

--> Atliq Hardware (fictitious corporation) is one of the major computer hardware manufacturers in India, with a strong presence in other nations.

--> Nevertheless, the management did note that they do not have sufficient insights to make prompt, wise, and data-informed judgments.

--> Plan to expand the data analytics team by adding junior data analysts.

--> To assess candidates, Data analytics director, Tony Sharma plans to conduct a SQL challenge to evaluate both tech and soft skills.

--> The company seeks insights for 10 ad hoc requests.

What?

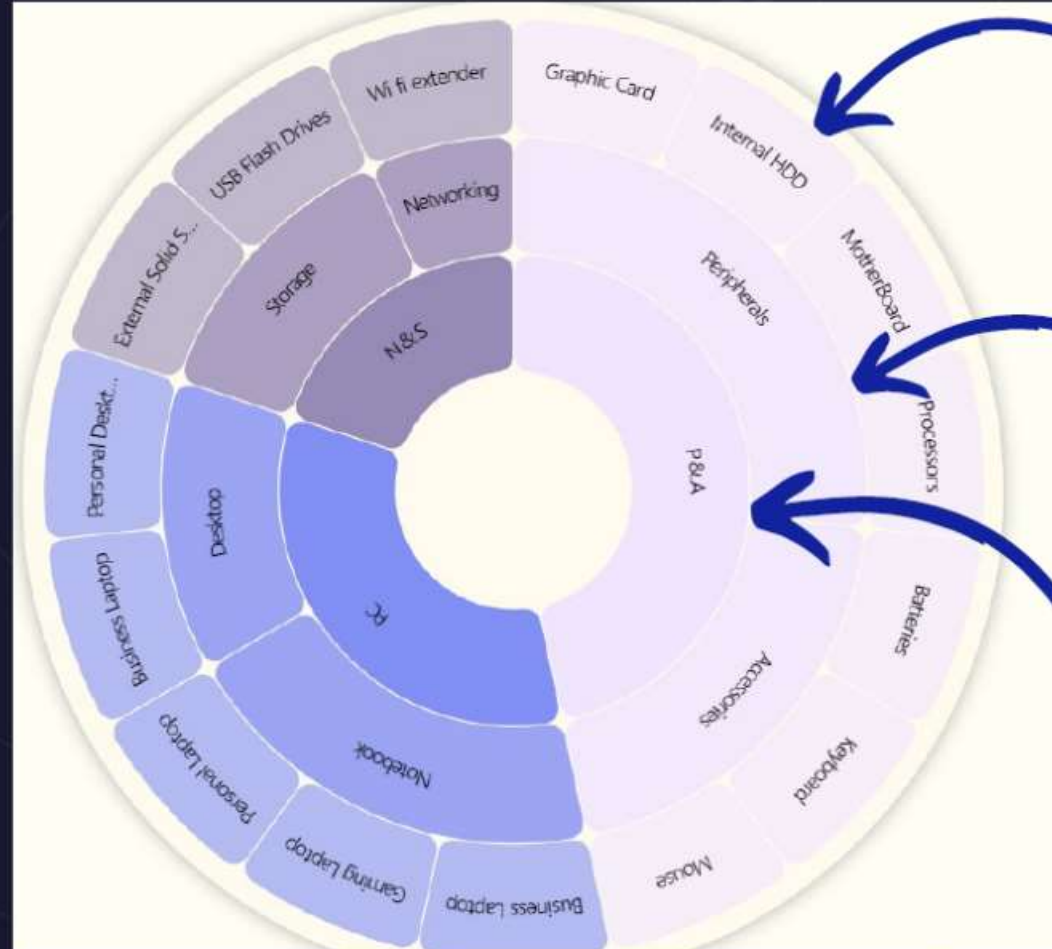
Company's Detail



Fiscal Year

SEPTEMBER 2019 - AUGUST 2020
FY 2020

SEPTEMBER 2020 - AUGUST 2021
FY 2021



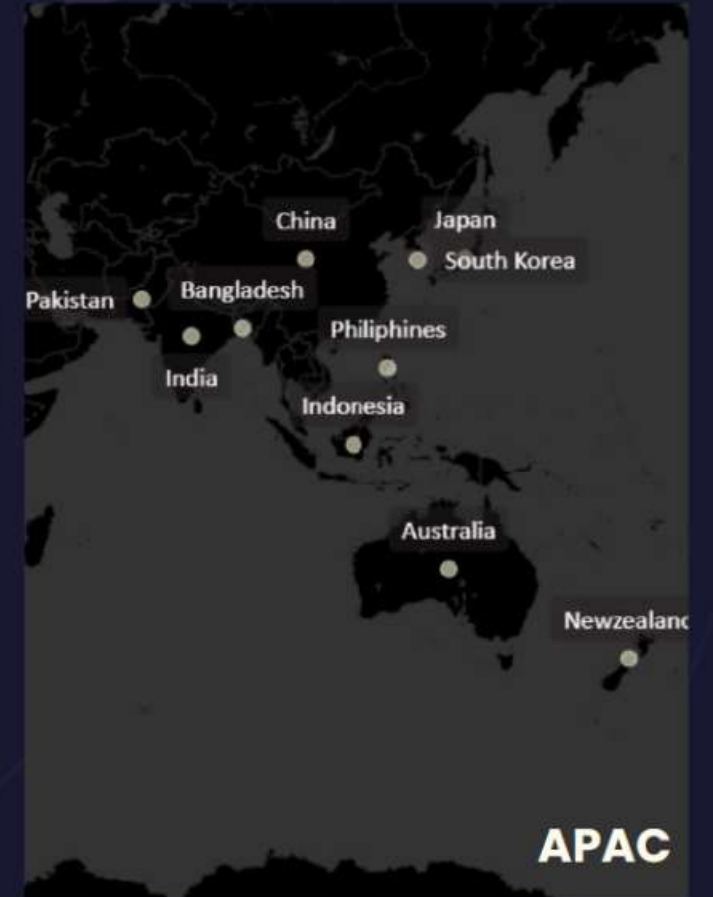
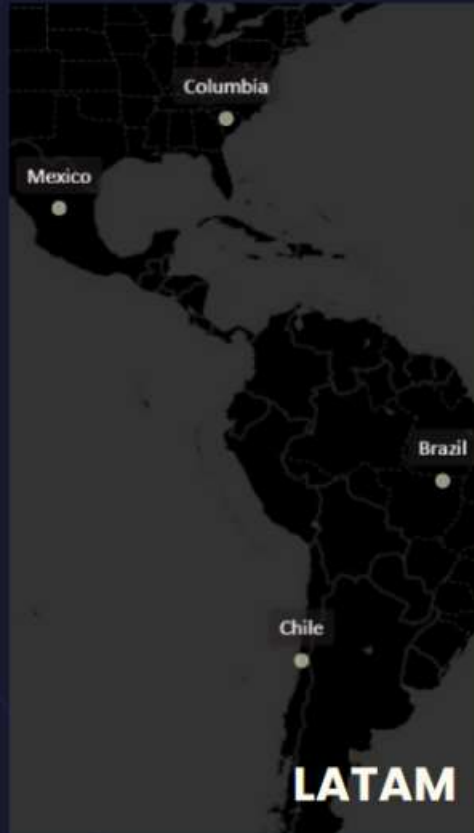
Outer Circle:
Category

Mid Circle:
Segment

Inner Circle:
Division



Company's Market



How?



Data, Request, and Tools

Power Bi & MySQL

For Analysis & Visualization

Codebasics SQL Challenge

Requests:

1. Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.
2. What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields,
unique_products_2020
unique_products_2021
percentage_chg
3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields,
segment
product_count
4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields,
segment
product_count_2020
product_count_2021
difference
5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields,
product_code
product
manufacturing_cost

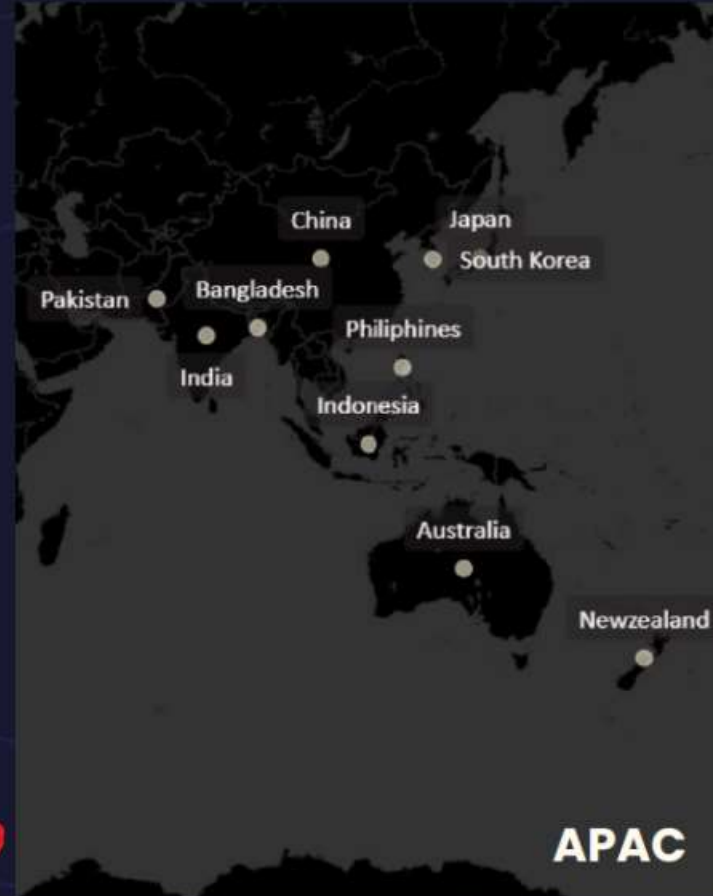
1. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,
customer_code
customer
average_discount_percentage
7. Get the complete report of the Gross sales amount for the customer "Atliq Exclusive" for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.
The final report contains these columns:
Month
Year
Gross sales Amount
8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity,
Quarter
total_sold_quantity
9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields,
channel
gross_sales_min
percentage
10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields,
division
product_code

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1. Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

market
India
Indonesia
Japan
Philippines
South Korea
Australia
Newzealand
Bangladesh



2. What is the percentage of unique product increase in 2021 vs. 2020?

The final output contains these fields,

unique_products_2020

unique_products_2021

percentage_chg

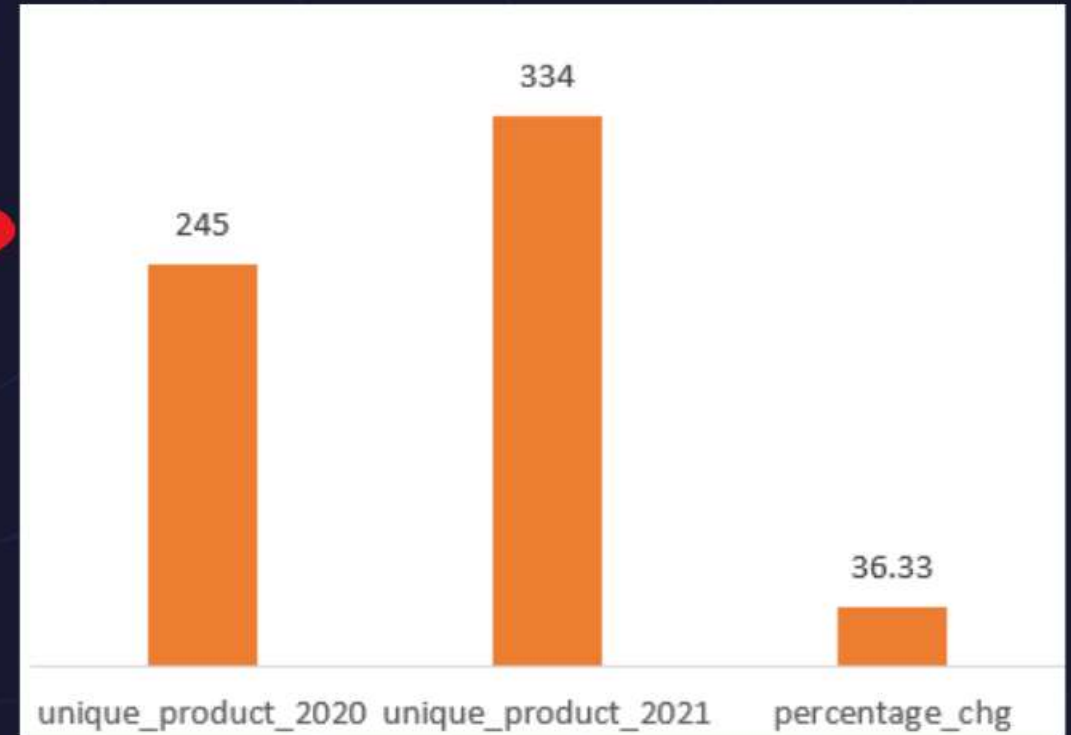
unique_product_2020

vs

unique_product_2021

unique_product_2020	unique_product_2021	percentage_chg
245	334	36.33

Percentage chg
36.33%



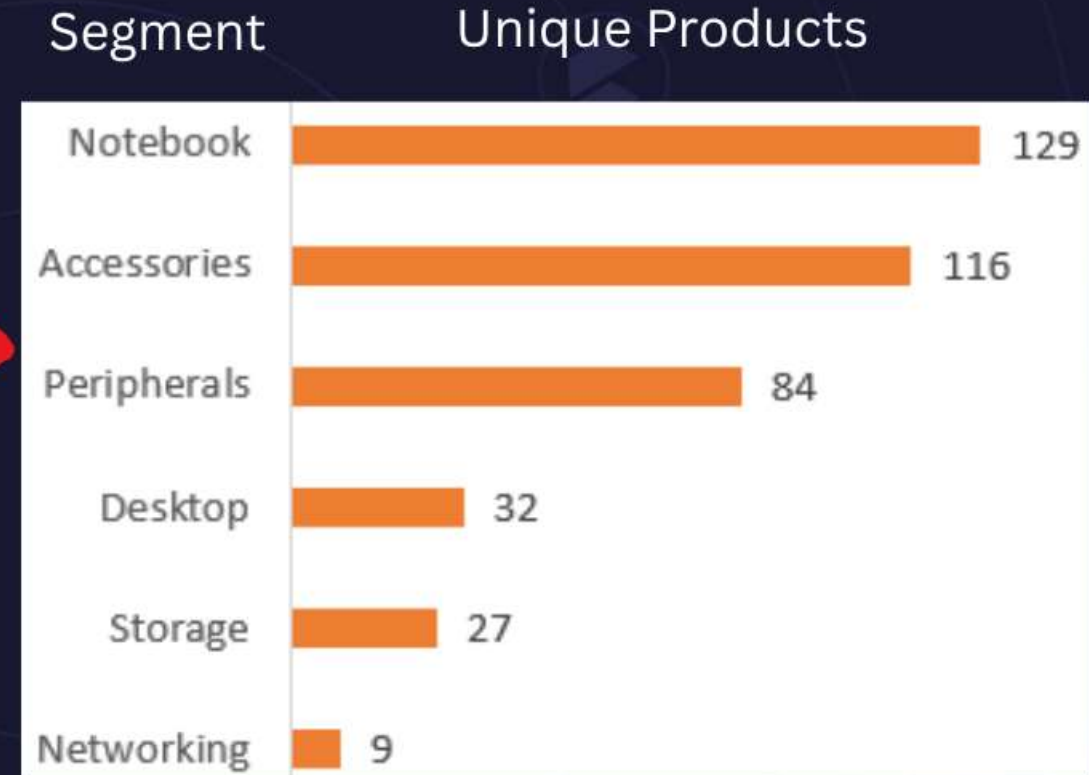
Insights:

Demand and production both increased.

3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields, Segment, Product_count



segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9



Insights:

- > Segments: notebooks, accessories, and peripherals are showing significant manufacturing growth as compared to desktops, storage, and networking.
- > Notebooks, accessories, and peripherals constitute 83% of the total manufactured product.

4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?
The final output contains these fields, segment product_count_2020 product_count_2021 difference.



Unique product difference per
segment from 2020 vs 2021

segment	product_count_2020	product_count_2021	difference
Accessories	69	103	34
Desktop	7	22	15
Networking	6	9	3
Notebook	92	108	16
Peripherals	59	75	16
Storage	12	17	5

segment	product_count_2020	product_count_2021	difference
Accessories	69	103	34 ↑
Notebook	92	108	16 ↑
Peripherals	59	75	16 ↑
Desktop	7	22	15 ↑
Storage	12	17	5 ↑
Networking	6	9	3 ↑
Total	245	334	89

Insights: --> Accessories had the largest increase in production
-->Storage and networking are experiencing slower production growth than other segments.

5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields,

product_code
product
manufacturing_cost

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920

240.54

max_manufacturing_cost

A6120110206
AQ HOME Allin1 Gen 2
Personal Desktop

0.89

min_manufacturing_cost

A2118150101
AQ Master wired x1 Ms
Mouse

Product & product code

Insights:

--> Mouse: AQ Master wired x1 Ms (Variant: Standard 1) has the lowest manufacturing cost.

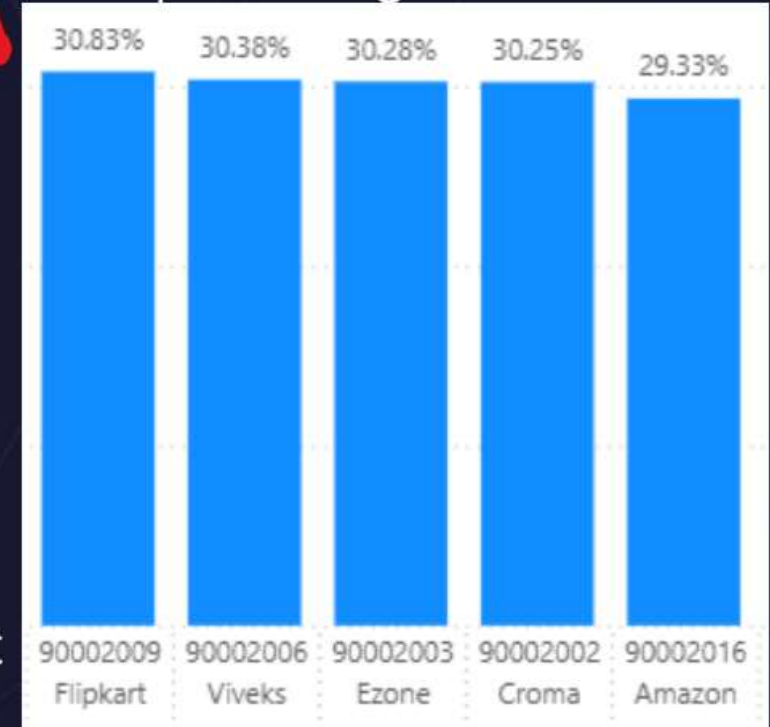
-->Personal Desktop: AQ Home Allin1 Gen2 (Variant: Plus 3) has the highest manufacturing cost.

6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market.
The final output contains these fields, customer_code customer average_discount_percentage.



customer_code	customer	average_discount_percentage
90002009	Flipkart	0.3083
90002006	Viveks	0.3038
90002003	Ezone	0.3028
90002002	Croma	0.3025
90002016	Amazon	0.2933

Top 5 indian customers with highest average discount percentage FY 2021



Insights:

- >The largest average pre-invoice discount was given to Flipkart
- >The least average pre-invoice discount was given to Amazon.

7. Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns: Month, Year, Gross sales Amount



Insights:

- >The lowest Gross sales total for both fiscal years is in March (2020) is 766976.45
- >The highest Gross sales total for both fiscal years is in November (2020) is 32247289.79

Month	fiscal_year	Gross_sales_amt
September(2019)	2020	9092670.34
October(2019)	2020	10378637.60
November(2019)	2020	15231894.97
December(2019)	2020	9755795.06
January(2020)	2020	9584951.94
February(2020)	2020	8083995.55
March(2020)	2020	766976.45
April(2020)	2020	800071.95
May(2020)	2020	1586964.48
June(2020)	2020	3429736.57
July(2020)	2020	5151815.40
August(2020)	2020	5638281.83
September(2020)	2021	19530271.30
October(2020)	2021	21016218.21
November(2020)	2021	32247289.79
December(2020)	2021	20409063.18
January(2021)	2021	19570701.71
February(2021)	2021	15986603.89
March(2021)	2021	19149624.92
April(2021)	2021	11483530.30
May(2021)	2021	19204309.41
June(2021)	2021	15457579.66
July(2021)	2021	19044968.82
August(2021)	2021	11324548.34

FY2020
79.5 M

FY2021
224.4 M



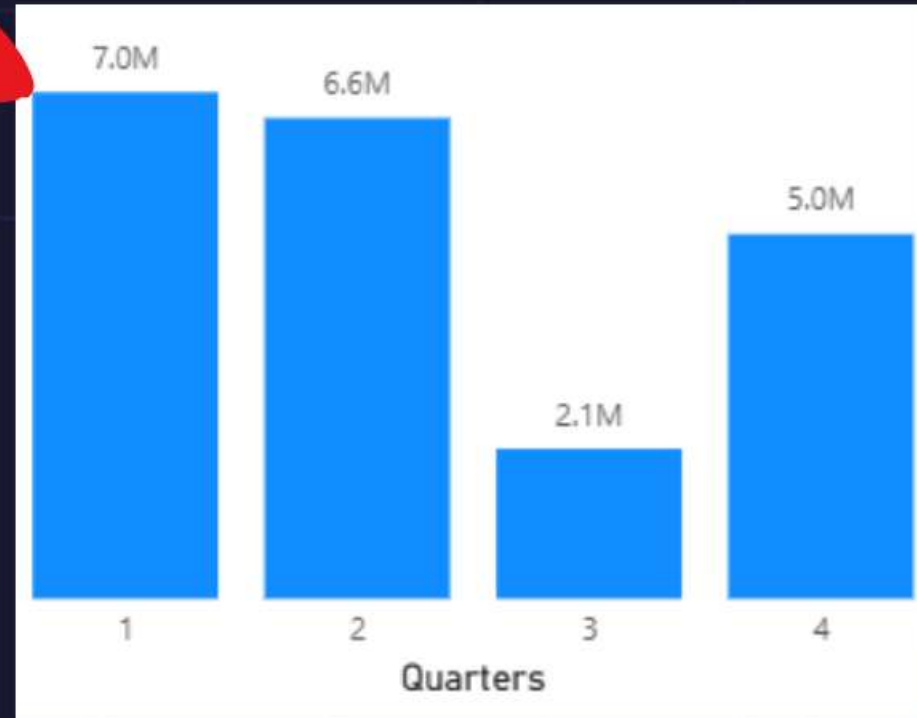
Reasons:

Covid-19
Global chip
shortage

8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity, Quarter , total_sold_quantity



Quarters	total_sold_qty
1	7005619
2	6649642
4	5042541
3	2075087



Insights:

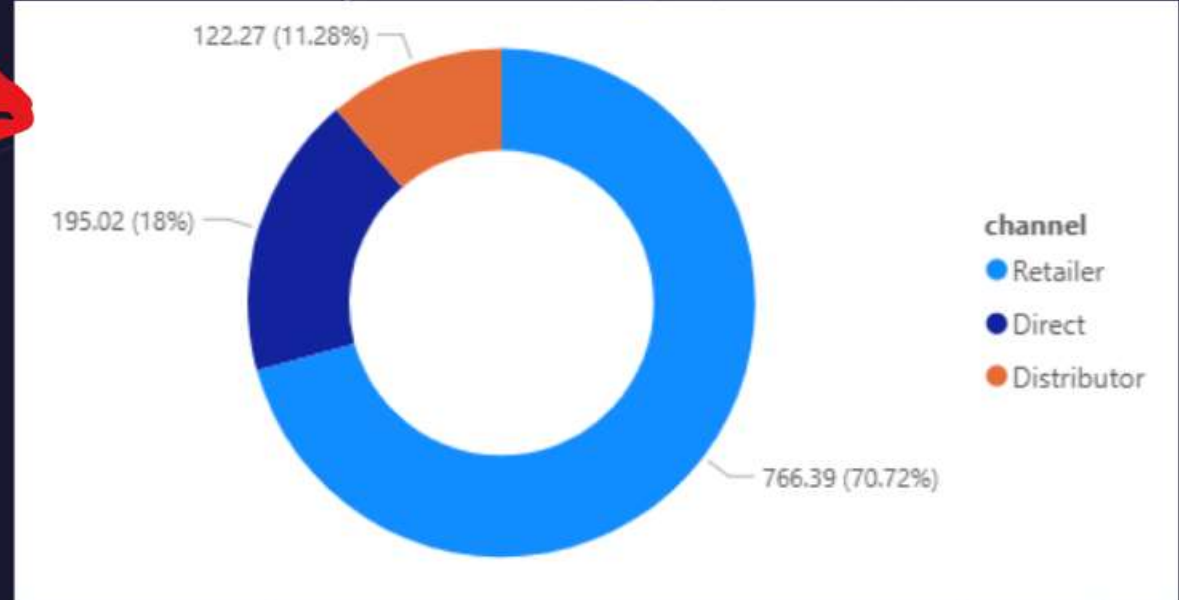
- > Quarter 1 of FY2020 saw the most units sold overall, while Quarter 3 had the fewest.
- > Quarter 1 accounts for approximately 34% of the total sold quantity for FY2020.

9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields, channel, gross_sales_mln, percentage



channel	gross_sales_mln	pct
Retailer	766.39	70.72%
Direct	195.02	18.00%
Distributor	122.27	11.28%

Gross sales and contribution percentages by channel for FY 2021



Insights:

Channel: "Retailer" helped bring maximum sales to the company with 73.22% as the contribution percentage.

Channel: "Distributor" makes the least contribution at a percentage of 11.31%.



10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields, division, product_code, product, total_sold_quantity, rank_order

division	product_code	product	total_sold_qty	Rank_order
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3

Insights:

Every division has a product with different variants that appears twice in the top three products by division list.

Division: N&S



Division: P&A



Division: PC





**END
THANK YOU!**

